

Transform based feature construction utilizing magnitude and phase for convolutional neural network in EEG signal classification

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Abstract—Extracting relevant feature and classification are significant in brain-computer interface (BCI) systems. Deep learning have achieved remarkable growth in many fields like speech recognition and computer vision. However, deep learning in biomedical field is yet to be fully utilized. In this paper, We propose a novel methodology for convolutional neural network (CNN) based motor imagery (MI) classification using new form of input. Continuous Wavelet Transform (CWT) is applied to the input Electroencephalography (EEG) signal to extract the features of MI. After transformation, we consider the real part and imaginary part of the transformed signal to exploit magnitude and phase information at the same time. This feature is fed to the CNN having one convolution layer, one max-pooling layer and one fully connected layer. The classification accuracy is tested on two public BCI datasets: BCI competition IV dataset IIb and BCI competition II dataset III. The proposed method shows increase in classification accuracy compared to other MI classification methods. The results show that the method using CNN with magnitude and phase based features can be better than other state-of-the-art approaches.

Index Terms—brain-computer interface (BCI), convolutional neural network (CNN), electroencephalography (EEG), motor imagery (MI)

I. INTRODUCTION

Brain-computer interfaces (BCIs) allow communication between the human brain and external devices by quantifying specific features of brain activity without requiring muscle control [1]–[6]. Among the non-invasive brain signal measuring methods, electroencephalography (EEG) is widely used due to its high temporal resolution, accessibility and low costs [7]–[10]. A lot of feature extraction algorithms have been suggested to decode EEG using time and frequency domain features [11]–[15].

Widely utilized methods like principal component analysis (PCA), independent component analysis (ICA), dynamic connectivity analysis, and frequency band analysis are used often to enhance the classification accuracy. The common spatial pattern (CSP) is well known method extracting discriminative features from EEG signals [11]–[15]. These discriminative features were used as an input to the classical supervised classifiers like linear discriminant analysis (LDA) [16], Bayesian classifier [17], and support vector machine (SVM) [18] for the classification of MI tasks.

Recently, deep learning approach have been rapidly improved reaching robust classification accuracy in pattern recog-

nition and speech recognition. Therefore, There have been attempts to apply deep learning methods into BCIs. In [19], deep belief network (DBN) structure is used for the classification of MI tasks. The performance of DBN surpassed SVM. In [20], deep neural network with denoising autoencoder is trained to classify MI tasks. Especially, Convolutional neural network (CNN) have been successfully utilized in MI classification. Several studies applied different decomposition method to make a image input to the CNN. In [21], fast Fourier transform (FFT) energy map is used to extract channel, frequency features of EEG signals. In [22], short time Fourier transform (STFT) is used to make EEG signals into 2D images combining time, frequency. STFT is used to get local section signal information. Then, the input image is fed to CNN combined with stacked automatic encoder (SAE) for classification of MI. In [23], wavelet transform (WT) is applied to convert EEG signals into time-frequency images which is used for the input of CNN. Method in [23] tried WT method to analyze non-stationary EEG signals and to avoid window size selection problem. However, deep learning methods in these studies utilized magnitude based features neglecting the phase information. Phase have additional information to MI and can be used as a discriminative features for MI. Therefore, features using both magnitude and phase information can be beneficial.

In this study, we present a new form of input which uses real part and imaginary part of the transformed EEG signals. Unlike methods in [21]–[23] where only the magnitude of the transformed EEG signal is considered, our method utilized real part and imaginary part explicitly to exploit the magnitude and phase information at the same time. Continuous wavelet transform (CWT) is applied in each electrodes. The real part of the transformed data is stacked to get real part image. In the same way, the imaginary part of the transformed data is stacked to get imaginary part image. Final image input is formed by combining real part image and imaginary part image. These magnitude and phase based features are fed to the CNN.

The classification accuracy of the proposed method is tested on the BCI competition IV dataset IIb, and BCI competition II dataset III. These datasets are publicly available datasets. The experiment results illustrate that using magnitude and phase based features for CNN showed higher classification accuracy than using a magnitude based features for CNN.

This paper is organized as follows. In section II, we formally introduce our input form and CNN architecture. In section III,

the BCI competition IV dataset IIb and BCI competition II dataset III are described. Section IV contains the simulation results. Finally, conclusions are made in Section V.

II. METHOD

A. Continuous wavelet transform

Consider EEG signals having C -electrodes $X \in \mathbb{R}^{C \times N}$, where N denotes the number of samples per electrode. EEG signals in each electrode $c = 1, 2, \dots, C$ is defined by $x_c[n]$, $n = 0, 1, \dots, N - 1$. The CWT of $x_c[n]$ is given by:

$$W_c(n, s) = \sum_{m=0}^{N-1} x_c[m] \psi^* \left[\frac{(m-n)\delta t}{s} \right] \quad (1)$$

$$= g_c(n, s) + j h_c(n, s)$$

where ψ is the mother wavelet function, δt is the sampling interval, $*$ indicates the complex conjugate, $s > 0$ is the wavelet scale, n is the translational value, $g_c(n, s)$ represents the real part of the c -th electrode, and $h_c(n, s)$ represents the imaginary part of the c -th electrode. To get the images of real part, $g_1, g_2, \dots, g_C$ are stacked vertically. Similarly, images of imaginary part is obtained by stacking $h_1, h_2, \dots, h_C$ vertically. Then images of real part and images of imaginary part are combined, forming final inputs with size $2 \times L_h \times L_w$ where L_h is the height of the final input and L_w is the width of the final input. Figure 1 shows the example images of real part and imaginary part when performing MI tasks.

B. Convolutional neural network architecture

Network architecture is composed of convolutional layer, max-pooling layer and fully connected layer as illustrated in Figure 2. Our input image which is described in section II, is formed by stacking each electrode's real part and imaginary part vertically. Therefore, capturing activation patterns in vertical location is crucial. To fully exploit our input data structure, convolution along the time axis using filters with same height of the input is used. Since our input form is 2 channel image of real part image and imaginary part image, we need to construct a convolution filters with 2 channels. For each channel, convolution is performed. Then the result of each channel is summed to produce output of the convolutional layer. Totally 30 filters with size $L_h \times 3$ are used to extract different feature maps. Convolution layer is followed by a Rectified linear Unit (ReLU) which gives non-linearity to the model. ReLU function is applied to the output of the convolution layer. Since we used filters with same height of the input, one dimensional output is obtained after the convolution layer. Therefore, in max-pooling layer, one dimensional pooling operation is performed. After the max-pooling layer, dropout [25] is applied. Dropout randomly drops units from the neural network during training to prevent units from co-adapting too much. Dropout rate is set to 0.5. The output of the max-pooling layer is flattened and fed to fully connected layer having two outputs representing left and right hand MI. Adam [26] is used as the optimization method.

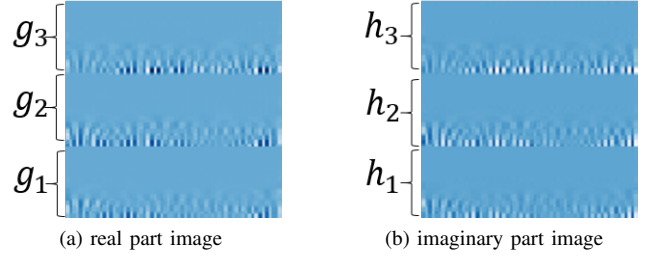


Fig. 1. Example image of CWT

TABLE I
DATASET ATTRIBUTES

Dataset	Subjects	Classes	Trials	Channels	Sampling rate
BCI competition IV dataset IIb	9	2	400	3	250
BCI competition II dataset III	1	2	280	3	128

III. DATA AND EXPERIMENTS

A. Data description

To evaluate the performance of our method, we used BCI competition IV dataset IIb [2] and BCI competition II dataset III [24].

BCI competition IV dataset IIb consists of EEG data from 9 normal right-handed subjects. The experiment includes three sessions in the training set. The EEG was bandpass-filtered between 0.5Hz and 100Hz. The first two sessions, At $t=2s$, a short warning beep was given. At $t=3s$, the arrow(left or right) was presented for 1.25s. Third online feedback session, Timing scheme is same with previous sessions. this session gives smiley feedback about correctness.

BCI competition II dataset III was recorded from a normal subject(female, 25y). The experiment consists of 280 trials of 9s length. The first 2s was quite, at $t=2s$ an acoustic stimulus indicates the beginning of the trial and a cross '+' was displayed for 1s. Then at $t=3s$, an arrow(left or right) was displayed as cue. Concurrently, the subject was asked to carry out the motor imagery task. The EEG was filtered between 0.5 and 30Hz. Attributes of the datasets are shown in Table 1.

B. Experiments

We use EEG signals from 0.25s to 3.25s after the cue. EEG signals are bandpass-filtered between 7Hz and 30Hz. 'cmor3-3' wavelet is used for the transformation.

1) *BCI competition IV dataset IIb*: Since the sampling rate is 250Hz, 750 time samples are obtained. After CWT, 32×750 complex-valued matrix is made. This process is repeated for 3 electrodes producing W_1, W_2 , and W_3 . Real part of the complex-valued matrix g_1, g_2 , and g_3 are stacked vertically forming 96×750 image. In the same way, imaginary part of the complex-valued matrix h_1, h_2 , and h_3 are stacked vertically forming 96×750 image. real part and imaginary part image is resized to 64×64 . Final input image form $2 \times 64 \times 64$ is obtained by combining real part image and imaginary part image. 30 filters with size 64×3 and stride of 1 is applied. In max-pooling layer, 1×10 filter and

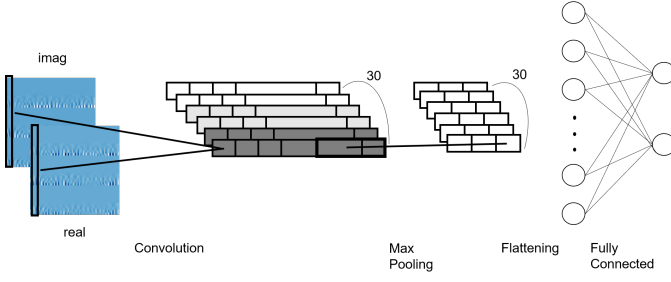


Fig. 2. CNN architecture of proposed method

stride of 10 is applied. The batch size of 50 and 300 epochs are used for the training of CNN.

2) *BCI competition II dataset III*: Since the sampling rate is 128Hz, 384 time samples are obtained. Reference [23] demonstrated that electrodes C3 and C4 are enough for the classification of MI. Hence, we used 2 electrodes C3 and C4 neglecting electrode Cz in this dataset. CWT is performed on C3 and C4 electrodes which result in W_1 , and W_2 . The real part g_1 and g_2 are stacked vertically. The imaginary part h_1 and h_2 are stacked vertically in the same way with real part image. Then real part image and imaginary part image are resized to 64×64 . The final input image form is $2 \times 64 \times 64$. 30 filters with size 64×3 and stride of 1 is applied. In max-pooling layer, 1×10 filter and stride of 10 is applied.

IV. RESULT AND DISCUSSION

In this section, we present the results of our experiments evaluated by using BCI competition IV dataset IIB and BCI competition II dataset III. First, we apply our method in BCI competition IV dataset IIB and its performance is compared with SVM and CNN-SAE methods. Second, we apply our method in BCI competition II dataset III. The performance of our method is compared with WT method.

A. Performance result in BCI competition IV dataset IIB

Each subject is trained and tested separately. 10×10 fold cross-validation accuracy is used to evaluate the performance. Table 2 shows the 10×10 fold cross-validation accuracy of our method and other methods. Proposed method shows average classification accuracy improvement of 5.9% and 0.7% compared to SVM and CNN-SAE method. Unlike magnitude based feature, Our method utilized both magnitude and phase information, thus improving classification accuracy.

B. Performance result in BCI competition II dataset III

The 5 fold cross-validation accuracy is used to evaluate the performance. The CNN is trained for 200 epochs with batch size 32.

Table 3 shows the 5 fold cross-validation accuracy of the proposed method and other method. As we can see in Table 3, our method shows higher classification accuracy in best case and in average case.

TABLE II
10 × 10 FOLD CROSS-VALIDATION CLASSIFICATION ACCURACY RESULT OF BCI COMPETITION IV DATASET IIB

Subjects	SVM	CNN-SAE	Proposed method
1	71.8	76.0	74.0
2	64.5	65.8	68.0
3	69.3	75.3	67.8
4	93.0	95.3	92.3
5	77.5	83.0	79.4
6	72.5	79.5	78.1
7	68.0	74.5	87.4
8	69.8	75.3	79.0
9	65.0	73.3	78.3
Average	72.4	77.6	78.3

TABLE III
5 FOLD CROSS-VALIDATION CLASSIFICATION ACCURACY RESULT OF BCI COMPETITION II DATASET III

	Worst	Best	Average
WT	87.5	92.8	89.6
Proposed method	85.7	94.6	90.0

V. CONCLUSION

We presented a novel MI classification methodology using CNN. In this study, we designed new form of input using real and imaginary part to utilize magnitude and phase information simultaneously. Simple CNN architecture was designed to prevent overfitting. Our architecture has one convolution layer, one max-pooling layer and fully connected layer. Result of our study shows that using magnitude and phase based input can improve the classification accuracy of MI.

ACKNOWLEDGMENT

This work was partly supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. 2017-0-00432, Development of non-invasive integrated BCI SW platform to control home appliances and external devices by user's thought via AR/VR interface) and Institute for Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. 2017-0-00451, Development of BCI based Brain and Cognitive Computing Technology for Recognizing User's Intentions using Deep Learning).

REFERENCES

- [1] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, J. R. Wolpaw, "BCI2000: A General-Purpose Brain-Computer Interface (BCI) System," *IEEE Trans. Biomed Eng.*, vol. 51, no. 6, pp. 1034-1043, Jun. 2004.
- [2] B. Blankertz, G. Dornhege, M. Krauledat, K. -R. Müller, and G. Curio, "The non-invasive Berlin brain-computer interface: Fast acquisition of effective performance in untrained subjects," *NeuroImage*, vol. 37, no. 2, pp. 539-550, 2007.
- [3] N.-S. Kwak, K.R. Müller, S.-W. Lee, "A convolutional neural network for steady state visual evoked potential classification under ambulatory environment," *PloS One*, vol. 12, no. 2, pp. e0172578, 2017.

- [4] S.-K. Yeom, S. Fazli, K.R. Müller, S.-W. Lee, “An efficient ERP-based brain-computer interface using random set presentation and face familiarity,” *PloS One.*, Vol. 9, No. 11, e111157, 2014.
- [5] M.-H. Lee, S. Fazli, J. Mehnert, S.-W. Lee, “Subject-dependent classification for robust idle state detection using multi-modal neuroimaging and data-fusion techniques in BCI,” *Pattern Recognit.*, Vol. 48, No. 8, pp. 2725-2737, 2015.
- [6] J.-H. Kim, F. Bießmann, S.-W. Lee, “Decoding three-dimensional trajectory of executed and imagined arm movements from electroencephalogram signals,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, Vol. 23, No. 5, pp. 867-876, 2014.
- [7] Y. Chen, A.D. Atnafu, I. Schlattner, W.T. Weldtsadik, M.-C. Roh, H.J. Kim, S.-W. Lee, B. Blankertz, S. Fazli, “A high-security EEG-based login system with RSVP stimuli and dry electrodes,” *IEEE Trans. nf. Forensics Secur.*, Vol. 11, No. 12, pp. 2635-2647, 2016.
- [8] I.-H. Kim, J.-W. Kim, S. Haufe, S.-W. Lee, “Detection of braking intention in diverse situations during simulated driving based on EEG feature combination,” *J. Neural Eng.*, Vol. 12, No. 1, 016001, 2014.
- [9] Y. Park, W. Chung, “BCI classification using locally generated CSP features,” *6th Int. Conf. Brain-Comput. Interface (BCI)*, 2018.
- [10] Y. Park, W. Chung, “Novel BCI classification method using cross-channel-region CSP features,” *6th Int. Conf. Brain-Comput. Interface (BCI)*, 2018.
- [11] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Müller, “Optimizing spatial filters for robust EEG single-trial analysis,” *IEEE Signal Process. Mag.*, vol. 25, no. 1, pp. 41-56, Dec. 2008.
- [12] H. Ramoser, J. Müller-Gerking, and G. Pfurtscheller, “Optimal spatial filtering of single trial EEG during imagined hand movement,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 14, no. 3, pp. 441-446, Dec. 2000.
- [13] Y. Park, W. Chung, “Selective Feature Generation Method Based on Time Domain Parameters and Correlation Coefficients for Filter-Bank-CSP BCI Systems,” *Sensors.*, vol. 19, no 17, 3769, 2019.
- [14] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, “Filter bank common spatial pattern algorithm on competition IV datasets 2a and 2b,” *Frontiers Neurosci.*, vol. 6, p. 39, Mar. 2012.
- [15] Y. Park and W. Chung, “Frequency-Optimized Local Region Common Spatial Pattern Approach for Motor Imagery Classification,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 7, pp. 1378-1388, 2019.
- [16] P. A. Lachenbruch, and M. Goldstein, “Discriminant analysis,” *Biometrics.*, vol. 35, no. 1, pp. 69-85, Mar. 1979.
- [17] P. Domingos, and M. Pazzani, “On the optimality of the simple bayesian classifier under zero-one loss,” *Machi. Learn.*, vol. 29, no. 2-3, pp. 103-130, Nov. 1997.
- [18] J.A.K. Suykens, and J. Vandewalle, “Least squares support vector machine classifiers,” *Neural Process Lett.*, vol. 9, no. 3, pp. 293-300, Jun. 1999.
- [19] X. An, D. Kuang, X. Guo, Y. Zhao, and L. He “A Deep Learning Method for Classification of EEG Data Based on Motor Imagery,” in *ICIC 2014*, pp. 203-210, 2014.
- [20] J. Li, A. Cichocki, “Deep Learning of Multifractal Attributes from Motor Imagery Induced EEG,” *Proc. ICONIP.*, pp. 503-510, 2014.
- [21] T. Uktveris, V. Jusas, “Application of Convolutional Neural Networks to Four-Class Motor Imagery Classification Problem,” *INF TECHNOL CONTROL.*, vol. 46. no. 2, pp. 260-273, 2017.
- [22] Y. R. Tabar and U. Halici, “A novel deep learning approach for classification of EEG motor imagery signals,” *J. Neural Eng.*, vol. 14, no. 1, p. 016003, Nov. 2017.
- [23] B. Xu, L. Zhang, A. Song, C. Wu, W. Li, D. Zhang, G. Xu, H. Li, and H. Zeng, “Wavelet transform time-frequency image and convolutional network-based motor imagery EEG classification,” *IEEE Access.*, vol. 7, , pp. 6084-6093, 2019.
- [24] B. Blankertz *et al.*, “The BCI competition 2003: Progress and perspectives in detection and discrimination of EEG single trials,” *IEEE Trans. Biomed. Eng.*, vol. 51, no. 6, pp. 1044-1051, 2004.
- [25] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, “Dropout: A simple way to prevent neural networks from overfitting,” *J. Mach. Learn. Res.*, vol. 15, pp. 1929-1958, 2014.
- [26] D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” in *Proc. 3rd Int. Conf. Learn. Represent. (ICLR)*, 2015.